

Advanced Statistical Methods II

Lecture 15: Quantile Regression I

Monitirtha Dey

BSDS Third Year Students

Indian Statistical Institute



First Semester, AY 2026–27

Outline of Today's Lecture

Motivation: why conditional quantiles?

From ordinary quantiles to regression quantiles

Interpretation of the quantile regression model

Estimation and computation

Why quantile regression is useful

Examples

What we will prove next time

After the mid-semester exam, we started robust regression methods.

- ▶ Lecture 13: motivation for robust statistics.
- ▶ Lecture 14: LAD regression, i.e. median regression.
- ▶ Lecture 15: from median regression to **quantile regression**.

OLS asks: *how does the conditional mean change?*
Quantile regression asks: *how does any chosen conditional quantile change?*

The limitation of a single mean line

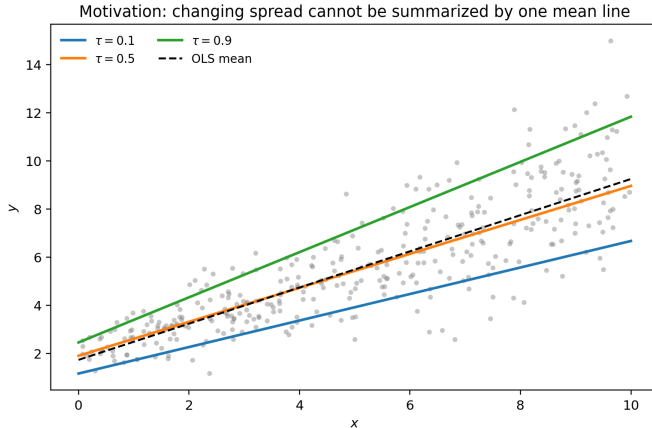


A regression line for the conditional mean can be very informative, but it can also hide important features.

- ▶ Does a predictor affect low responses and high responses in the same way?
- ▶ Does variability increase with the predictor?
- ▶ Are the tails more important than the centre?
- ▶ Is the response distribution skewed?

If the lower tail, median, and upper tail move differently, then one mean regression line is not enough.

One mean line misses changing spread



- ▶ The dashed black line summarizes the conditional mean.
- ▶ The conditional spread increases with x .
- ▶ Quantile lines reveal more of the conditional distribution.

Real data motivation: Engel's law

The Engel data record household income and food expenditure.

- ▶ A mean model estimates average food expenditure at a given income.
- ▶ But lower-spending and higher-spending households may respond differently to income.
- ▶ Quantile regression allows income effects to vary across the conditional distribution.

Question of interest

Does additional income affect lower, median, and upper food expenditure households equally?

This is a standard illustration in the quantile regression literature; see, for example, [Koenker \(2005\)](#) and [Koenker and Hallock \(2001\)](#).

Engel data: what is this dataset?



- ▶ The Engel dataset is a classical household-expenditure dataset used to illustrate Engel's law.
- ▶ It contains $n = 235$ observations on household income and food expenditure.
- ▶ In the dataset used here:

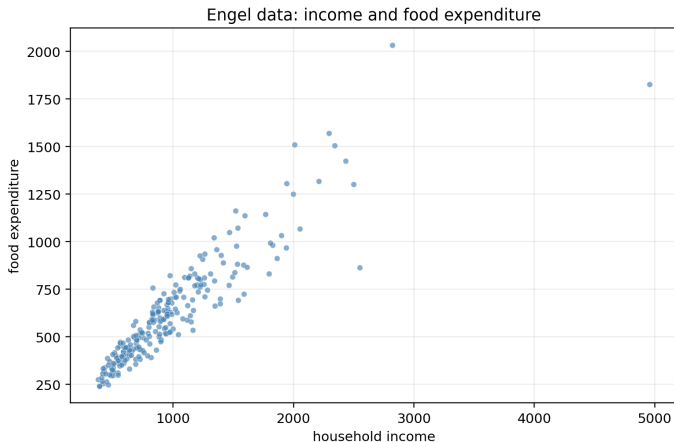
$$\min(\text{income}) = 377.06, \quad \max(\text{income}) = 4957.81,$$

$$\min(\text{food expenditure}) = 242.32, \quad \max(\text{food expenditure}) = 2032.68.$$

- ▶ Historically, Engel's law says that as income rises, the share of income spent on food tends to fall; the absolute amount spent on food can still increase.

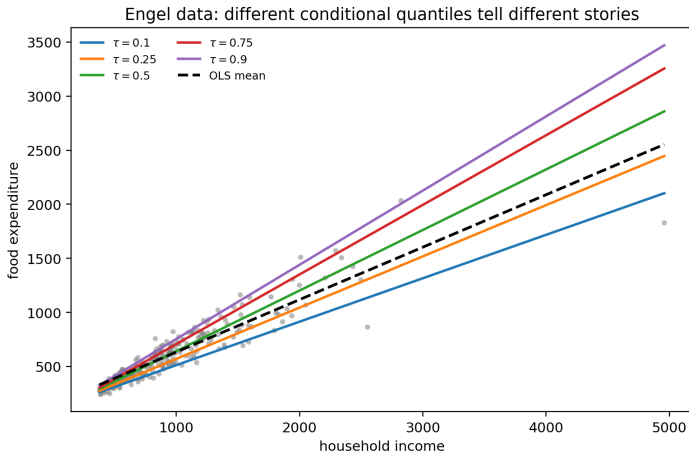
This dataset is a standard example in quantile regression; see [Koenker \(2005\)](#). The economic idea traces back to Engel's nineteenth-century work on household budgets ([Engel, 1857](#)).

Engel data: scatterplot



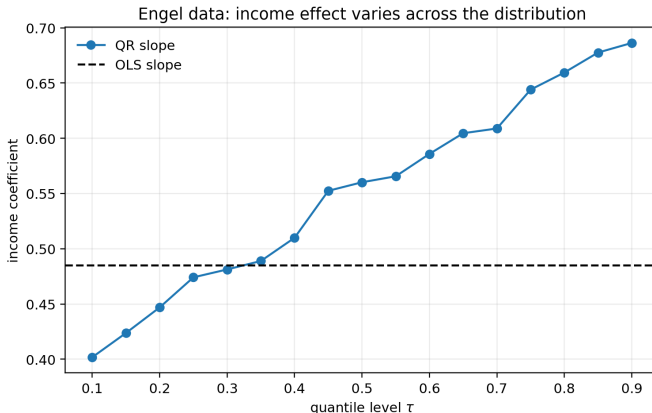
There is a positive relationship between income and food expenditure, but the spread and upper tail are not well summarized by one mean line.

Engel data: OLS and several quantile lines



The conditional quantile lines are not merely vertical shifts of the OLS line; the slopes can differ across quantiles.

Engel data: slope as a function of quantile level



- ▶ The OLS slope is one number.
- ▶ The quantile regression slope is indexed by τ .
- ▶ This can show how covariate effects change from lower to upper conditional quantiles.

Examples where quantiles are natural

1. **Economics:** wage or expenditure studies. Effects may differ for low-income and high-income groups ([Koenker and Hallock, 2001](#)).
2. **Public health / health services:** medical costs, hospital stay, and treatment outcomes; tail behaviour can matter more than the mean ([Cook and Manning, 2013](#)).
3. **Psychiatry:** neuropsychological performance or symptom scores may have covariate effects that differ across lower and upper quantiles ([Shankar et al., 2025](#)).
4. **Environment:** pollution, rainfall, cyclone intensity, and temperature extremes; upper quantiles may be the main scientific focus ([Miller and Neocleous, 2020](#), p. 53–54; PDF pp. 56–57).
5. **Education:** test scores. Predictors may affect weaker and stronger students differently ([Petscher and Logan, 2014](#)).
6. **Finance:** risk management. Lower-tail quantiles connect to value-at-risk type ideas ([Koenker, 2005](#)).

Quantile regression is useful when the entire conditional distribution is scientifically meaningful, not just its mean.

Sample median as an optimization problem



For observations $y_1, \dots, y_n$, the sample median minimizes

$$\sum_{i=1}^n |y_i - a|.$$

This was the starting point of LAD regression.

- ▶ OLS location estimate: minimizes squared deviations.
- ▶ LAD location estimate: minimizes absolute deviations.
- ▶ Quantile location estimate: minimizes **asymmetrically weighted absolute deviations**.

The median treats overprediction and underprediction symmetrically.

What if we want the 0.9 quantile instead?

The check loss function



For $0 < \tau < 1$, define the **check loss**

$$\rho_{\tau}(u) = \begin{cases} \tau u, & u \geq 0, \\ (\tau - 1)u, & u < 0. \end{cases}$$

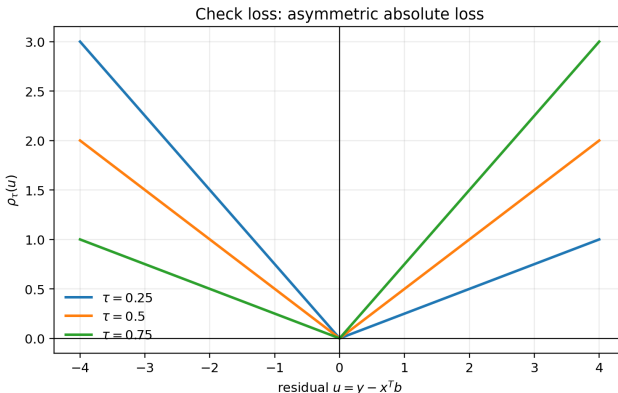
Equivalently,

$$\rho_{\tau}(u) = u\{\tau - \mathbb{I}(u < 0)\}.$$

- ▶ If $\tau = 1/2$, then $\rho_{1/2}(u) = \frac{1}{2}|u|$.
- ▶ If τ is large, positive residuals are penalized more than negative residuals.
- ▶ This asymmetry forces the fitted value upward toward an upper quantile.

This is the basic objective used in quantile regression ([Koenker and Bassett, 1978](#); [Koenker, 2005](#); [Miller and Neocleous, 2020](#)).

Check loss: picture



The check loss is convex but not differentiable at zero. This is why quantile regression has a different computational flavour from least squares.

Quantiles as minimizers of expected check loss



Let Y be a random variable with distribution function F . A τ th quantile is any value q_τ satisfying

$$F(q_\tau^-) \leq \tau \leq F(q_\tau).$$

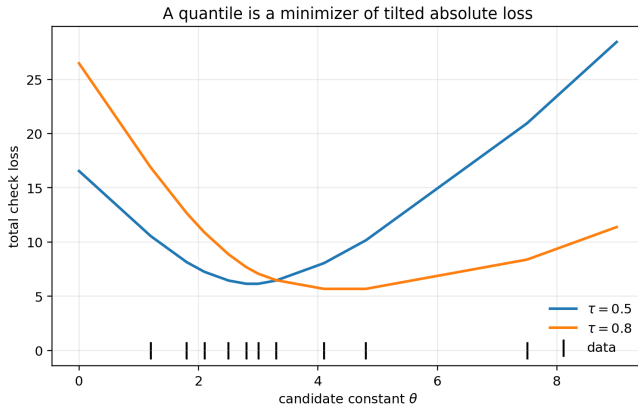
A key characterization is

$$q_\tau \in \arg \min_a \mathbb{E}\{\rho_\tau(Y - a)\}.$$

- ▶ $\tau = 0.5$: median.
- ▶ $\tau = 0.25$: lower quartile.
- ▶ $\tau = 0.90$: upper tail quantile.

Quantile regression is built by replacing the scalar a with a regression function.

Sample quantiles through tilted loss



- ▶ The minimum of the symmetric loss gives a median-type summary.
- ▶ Changing τ tilts the loss and shifts the minimizer.

From LAD to quantile regression



Recall LAD regression:

$$\hat{\beta}_{\text{LAD}} = \arg \min_{\mathbf{b} \in \mathbb{R}^p} \sum_{i=1}^n |y_i - \mathbf{x}_i^\top \mathbf{b}|.$$

Quantile regression replaces $|u|$ by $\rho_\tau(u)$:

$$\hat{\beta}(\tau) = \arg \min_{\mathbf{b} \in \mathbb{R}^p} \sum_{i=1}^n \rho_\tau(y_i - \mathbf{x}_i^\top \mathbf{b}).$$

LAD is the special case $\tau = 0.5$ of quantile regression.

In linear quantile regression, we model

$$Q_Y(\tau \mid \mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta}(\tau).$$

- Quantile regression is a way to model response quantiles, not only the mean (Miller and Neocleous, 2020).

$$Q_Y(\tau \mid x) = \beta_0(\tau) + \beta_1(\tau)x.$$

Interpretation

If $\hat{\beta}_1(0.9) = 0.68$ in the Engel example, then among households at the upper conditional food-expenditure quantile, a one-unit increase in income is associated with about 0.68 additional units of food expenditure.

This is a statement about conditional quantiles of the response distribution.

What does it mean when slopes differ?



If

$$\beta_1(0.1) < \beta_1(0.5) < \beta_1(0.9),$$

then the effect of x is stronger in the upper tail than in the lower tail.

- ▶ The relationship is not fully summarized by a single mean slope.
- ▶ The conditional distribution changes shape as x changes.
- ▶ This may indicate heteroscedasticity, skewness changes, or tail-specific covariate effects.

Quantile regression lets us ask: *which part of the response distribution is affected?*

Comparison with OLS and LAD



Method	Objective	Target
OLS	$\sum_i (y_i - \mathbf{x}_i^\top \mathbf{b})^2$	conditional mean
LAD	$\sum_i y_i - \mathbf{x}_i^\top \mathbf{b} $	conditional median
QR	$\sum_i \rho_\tau(y_i - \mathbf{x}_i^\top \mathbf{b})$	conditional τ -quantile

- ▶ OLS is efficient under homoscedastic normal errors.
- ▶ LAD is robust to large vertical outliers and targets the median.
- ▶ QR generalizes LAD to any chosen part of the conditional distribution.

What is new relative to LAD?



LAD answers one question:

$$Q_Y(0.5 \mid x) = x^\top \beta(0.5).$$

Quantile regression answers many questions:

$$Q_Y(\tau \mid x) = x^\top \beta(\tau), \quad 0 < \tau < 1.$$

- ▶ The median is only the centre of the conditional distribution.
- ▶ Low quantiles are useful for lower-tail behaviour.
- ▶ High quantiles are useful for upper-tail behaviour.
- ▶ Fitting a grid of τ values gives a distributional view.

The optimization problem



For fixed $\tau \in (0, 1)$, the linear quantile regression estimator is

$$\hat{\beta}(\tau) = \arg \min_{\mathbf{b} \in \mathbb{R}^p} \sum_{i=1}^n \rho_{\tau}(y_i - \mathbf{x}_i^{\top} \mathbf{b}).$$

- ▶ The objective is convex.
- ▶ It is piecewise linear.
- ▶ Not differentiable at residuals equal to zero.
- ▶ It can be formulated as a linear programming problem.

The original regression quantile formulation is due to [Koenker and Bassett \(1978\)](#).

Linear programming form



Write residuals as the difference of two nonnegative parts:

$$y_i - \mathbf{x}_i^\top \mathbf{b} = u_i - v_i, \quad u_i \geq 0, \quad v_i \geq 0.$$

Then

$$\rho_\tau(y_i - \mathbf{x}_i^\top \mathbf{b}) = \tau u_i + (1 - \tau)v_i.$$

So the quantile regression problem is equivalent to

$$\min_{\mathbf{b}, u, v} \sum_{i=1}^n \{\tau u_i + (1 - \tau)v_i\}$$

subject to

$$y_i - \mathbf{x}_i^\top \mathbf{b} = u_i - v_i, \quad u_i, v_i \geq 0.$$

This is why quantile regression historically developed through optimization and linear programming.

```
library(quantreg)

# median regression
fit50 <- rq(foodexp ~ income, data = engel, tau = 0.50)

# upper-tail regression
fit90 <- rq(foodexp ~ income, data = engel, tau = 0.90)

# several quantile levels
taus <- c(0.10, 0.25, 0.50, 0.75, 0.90)
fit_all <- rq(foodexp ~ income, data = engel, tau = taus)
summary(fit_all)
```

- ◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡|≡ ↺ 🔍 ↻

One small numerical interpretation



For the Engel data, an OLS fit gives an income slope around 0.485 in the analysis used for the plot.

The quantile regression slopes were approximately

$$\hat{\beta}_1(0.10) = 0.402, \quad \hat{\beta}_1(0.50) = 0.560, \quad \hat{\beta}_1(0.90) = 0.686.$$

- ▶ The estimated slope is larger at higher conditional quantiles.
- ▶ A mean model would hide this variation.
- ▶ A median model would still give only one central conditional summary.

Advantage 1: distributional heterogeneity



Quantile regression can reveal heterogeneity that mean regression cannot.

- ▶ The coefficient $\beta_j(\tau)$ may change with τ .
- ▶ This can detect different covariate effects in the lower, middle, and upper tails.
- ▶ This is especially useful when errors are heteroscedastic or skewed.

A covariate can have little effect on the mean but strong effect on the upper tail or lower tail.

Advantage 2: robustness compared with OLS



Like LAD, quantile regression is based on an absolute-loss-type criterion rather than squared loss.

- ▶ Large residuals are not squared.
- ▶ Median regression is resistant to vertical outliers in the response.
- ▶ Other quantiles are also less dominated by extreme values than OLS.

Quantile regression is not automatically immune to high-leverage covariate outliers. Robustness still depends on the design points.

$$Q_Y(\tau \mid X = x) = x^\top \beta(\tau),$$

- ▶ No normal error assumption is needed.
- ▶ No constant conditional variance assumption is needed.
- ▶ No parametric family such as Gaussian, Gamma, Weibull, etc. is required for the whole conditional distribution.
- ▶ The key modelling assumption is instead about the **conditional quantile function** at the chosen τ .

Miller and Neocleous write that quantile regression “does not make any distributional assumptions” and is therefore distribution-free; see [Miller and Neocleous \(2020, p. 54; PDF p. 57\)](#). This should not be confused with saying that the linear quantile model is automatically correct.

Advantage 4: meaningful tail questions



Many scientific and policy questions are naturally tail questions.

Examples

- ▶ What predicts very high medical expenditure?
- ▶ What predicts unusually low crop yield?
- ▶ What predicts very poor educational outcomes?
- ▶ What predicts very high pollution levels?

OLS answers a central-average question; quantile regression can answer lower-tail and upper-tail questions directly.

Modelling Extreme Events

- ▶ Results depend on the chosen τ values.
- ▶ Extreme quantiles need larger samples.
- ▶ Different fitted quantile curves can cross if fitted separately.
- ▶ Interpretation is conditional-quantile interpretation, not individual trajectory interpretation.
- ▶ Inference is more delicate than ordinary least squares.

Quantile regression should not be used merely to draw many lines.
The selected quantiles should match the scientific question.

Caution 1: the choice of τ matters



A quantile regression coefficient is not one object; it is a function of the quantile level:

$$\beta(0.10), \quad \beta(0.25), \quad \beta(0.50), \quad \beta(0.75), \quad \beta(0.90), \dots$$

- ▶ Different τ values answer different scientific questions.
- ▶ Reporting only selected quantiles can miss features between them.
- ▶ The chosen quantiles should be motivated by the application: lower tail, centre, upper tail, or a full grid.

Caution 2: why extreme quantiles need larger samples



For an ordinary sample quantile $\hat{q}_\tau$, under regularity conditions,

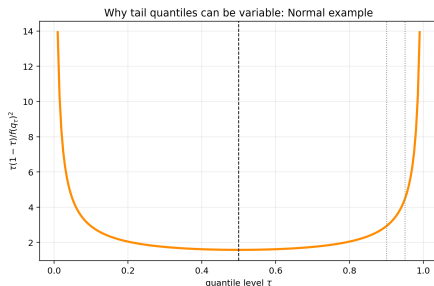
$$\sqrt{n}(\hat{q}_\tau - q_\tau) \xrightarrow{d} N\left(0, \frac{\tau(1-\tau)}{f(q_\tau)^2}\right).$$

Thus,

$$\text{Var}(\hat{q}_\tau) \approx \frac{\tau(1-\tau)}{nf(q_\tau)^2}.$$

- ▶ It is **not** enough to look only at $\tau(1-\tau)$.
- ▶ In the tails, the density $f(q_\tau)$ is often small.
- ▶ A small $f(q_\tau)$ inflates the variance through $1/f(q_\tau)^2$.

Extreme quantiles: visual and finite-sample intuition



If $n = 100$:

- ▶ the median has about 50 observations on each side;
- ▶ the 0.90 quantile has about 10 observations above it;
- ▶ the 0.95 quantile has about 5 observations above it;
- ▶ the 0.99 quantile has about 1 observation above it.

Tail quantiles are data-hungry because only a small fraction of observations informs that part of the distribution, and the conditional density near the tail quantile can be small.

Caution 3: conditional quantiles are not individual trajectories



Suppose

$$Q_Y(0.90 \mid X = x) = \beta_0(0.90) + \beta_1(0.90)x.$$

This compares the **90th percentile of Y among individuals with $X = x$** across different values of x .

It does not mean that a particular person who is currently at the 90th percentile will follow that same fitted curve as x changes.

Example: if Y is salary and X is years of experience, then

$$Q_Y(0.90 \mid X = 5) \quad \text{and} \quad Q_Y(0.90 \mid X = 10)$$

refer to the 90th percentile in two conditional populations. They need not refer to the same individuals.

Conditional distribution versus individual movement



Correct interpretation

At each fixed x , the fitted line estimates the τ th quantile of the conditional distribution of $Y \mid X = x$.

Incorrect interpretation: the curve is the life path of a “ τ th quantile individual.”

For example, a 90th-percentile height curve by age is a sequence of population quantiles. A child can move from the 75th percentile to the 90th percentile over time.

To interpret quantile curves as individual trajectories requires additional longitudinal or structural assumptions, such as rank preservation.

Caution 4: quantile crossing



For a true conditional distribution,

$$Q_Y(0.25 | x) \leq Q_Y(0.50 | x) \leq Q_Y(0.75 | x)$$

for every x .

But if we fit the three quantile regressions separately, finite-sample noise can lead to

$$\hat{Q}_Y(0.75 | x) < \hat{Q}_Y(0.50 | x)$$

for some x .

- ▶ Crossing is impossible for true quantiles.
- ▶ It can occur in separately fitted quantile curves.
- ▶ Joint estimation or rearrangement methods can be used when monotonicity is important.

Caution 5: inference is more delicate than OLS



OLS has a smooth quadratic objective:

$$\sum_i (y_i - x_i^\top b)^2.$$

Quantile regression uses the non-smooth check loss:

$$\sum_i \rho_\tau(y_i - x_i^\top b).$$

- ▶ Standard errors depend on density/sparsity near the target quantile.
- ▶ Bootstrap methods are often used in applications.
- ▶ Simultaneous inference across many quantiles is more complex than inference at one fixed τ .

QR is conceptually simple, but careful inference requires attention to the target quantile, density near that quantile, and the sampling design.

A simple diagnostic



When looking at a regression dataset, ask:

1. Does the scatter around the regression line increase or decrease with x ?
2. Do low and high outcomes seem to move differently with x ?
3. Is the response skewed for some values of x ?
4. Are tail outcomes scientifically important?

If the answer to any of these is yes, quantile regression is worth considering.

Worked interpretation exercise



Suppose we fit

$$Q_Y(\tau | x) = \beta_0(\tau) + \beta_1(\tau)x.$$

The estimated slopes are

$$\hat{\beta}_1(0.1) = 0.2, \quad \hat{\beta}_1(0.5) = 0.5, \quad \hat{\beta}_1(0.9) = 1.1.$$

1. What does this suggest about the conditional distribution of Y ?
2. Would one OLS line summarize this pattern well?
3. Which quantile would you report if the main concern is unusually large responses?

1. The effect of x is stronger in the upper tail than in the lower tail.
2. One OLS line would miss the changing slope across quantiles.
3. If unusually large responses are the focus, a high quantile such as $\tau = 0.9$ is appropriate.

Quantile regression should not be viewed as a *robust* trick only; it also gives us one way to study conditional distributional structure.

Lecture 15 versus Lecture 16



In this lecture, the emphasis is on motivation, definitions, interpretation, and examples.

In Lecture 16, we will discuss more theoretical details.

- ▶ Why the check loss targets quantiles.
- ▶ First-order/subgradient conditions.
- ▶ Basic equivariance properties.
- ▶ Consistency and asymptotic normality.
- ▶ Inference and practical implementation issues.

Key takeaway messages



1. OLS describes the conditional mean.
2. LAD describes the conditional median.
3. Quantile regression describes any chosen conditional quantile.
4. Different quantiles can have different regression slopes.
5. This makes quantile regression valuable under heteroscedasticity, skewness, and tail-specific scientific questions.

A good mental model: *Quantile regression replaces a single regression line by a family of distribution-sensitive regression lines.*

- ◀ ◻ ▶ ◀ ◻ ▶ ◀ ≡ ▶ ◀ ≡ ▶ ≡|≡ ↺ 🔍 ↻

- Shankar, R. G., Kandavel, T., Kashyap, H., and Reddy, Y. C. J. (2025). A case study on using quantile regression in psychiatry research. *Frontiers in Psychiatry*, 16, Article 1632001. doi:10.3389/fpsy.2025.1632001.
- Petscher, Y. and Logan, J. A. R. (2014). Quantile regression in the study of developmental sciences. *Child Development*, 85(3), 861–881.